Experiment - 7

Download and install nmap. Use it with different options to scan open ports, perform OS fingerprinting, do a ping scan, tcp port scan, udp port scan etc

Aim: To Understand the working of nmap using different options to scan open ports, perform OS fingerprinting, do a ping scan, tcp port scan, udp port scan, xmas scan etc.

Theory : Nmap (Network Mapper) is a security scanner used to discover hosts and services on a computer network, thus creating a "map" of the network. To accomplish its goal, Nmap sends specially crafted packets to the target host and then analyzes the responses. Unlike many simple port scanners that just send packets at some predefined constant rate, Nmap accounts for the network conditions (latency fluctuations, network congestion, the target interference with the scan) during the run. Also, owing to the large and active user community providing feedback and contributing to its features, Nmap has been able to extend its discovery capabilities beyond simply figuring out whether a host is up or down and which ports are open and closed; it can determine the operating system of the target, names and versions of the listening services, estimated uptime, type of device and presence of a firewall.

Nmap features include:

• **Host Discovery** – Identifying hosts on a network. For example, listing the hosts which respond to pings or have a particular port open.

• **Port Scanning** – Enumerating the open ports on one or more target hosts.

• **Version Detection** – Interrogating listening network services listening on remote devices to determine the application name and version number.

• **OS Detection** – Remotely determining the operating system and some hardware characteristics of network devices.

Installing Nmap

sudo apt-get install nmap

1: Scan a single host or an IP address

nmap 192.168.3.145

nmap mu.ac.in

Scan multiple IP address

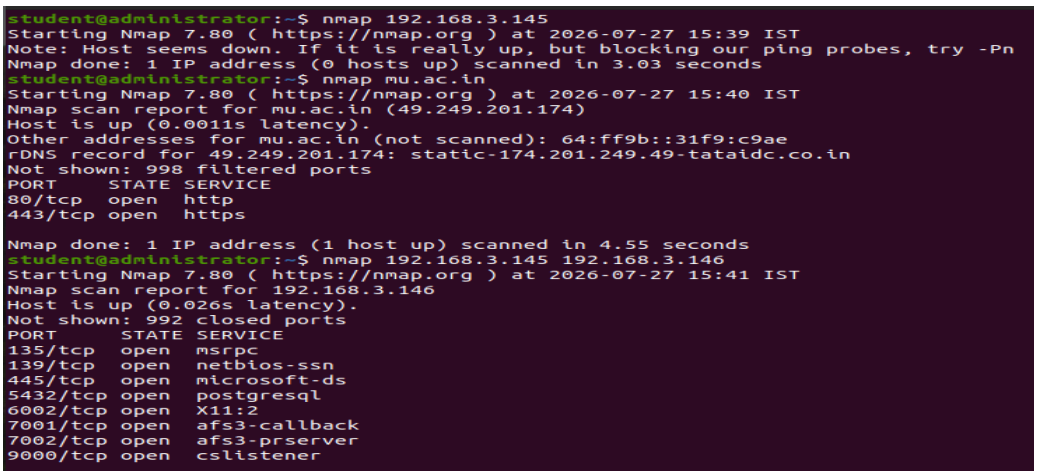
nmap 192.168.3.145 192.168.3.146

You can scan a range of IP address too:

nmap 192.168.3.145-155

You can scan a range of IP address using a wildcard:

nmap 192.168.3.*



```
student@administrator:~$ nmap 192.168.3.145-155
Starting Nmap 7.80 ( https://nmap.org ) at 2026-07-27 15:42 IST
Nmap scan report for 192.168.3.146
Host is up (0.0017s latency).
Not shown: 992 closed ports
PORT     STATE SERVICE
135/tcp  open  msrpc
139/tcp  open  netbios-ssn
445/tcp  open  microsoft-ds
5432/tcp open  postgresql
6002/tcp open  X11:2
7001/tcp open  afs3-callback
7002/tcp open  afs3-prserver
9000/tcp open  cslistener

Nmap scan report for 192.168.3.153
Host is up (0.00030s latency).
Not shown: 968 filtered ports, 30 closed ports
PORT   STATE SERVICE
21/tcp open  ftp
80/tcp open  http

Nmap scan report for 192.168.3.155
Host is up (0.00017s latency).
All 1000 scanned ports on 192.168.3.155 are closed

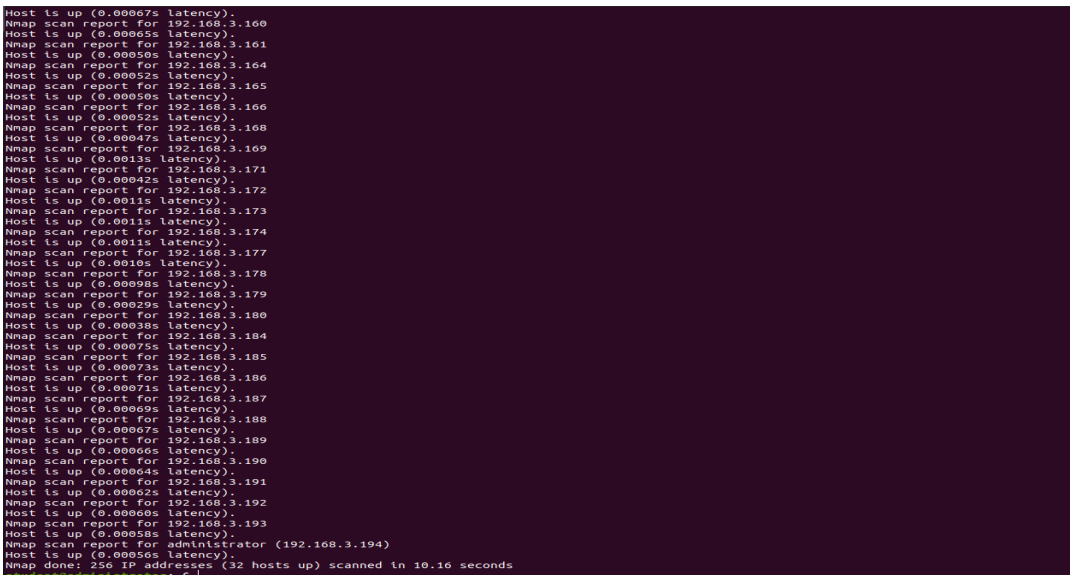
Nmap done: 11 IP addresses (3 hosts up) scanned in 6.70 seconds
student@administrator:~$ nmap 192.168.3.*
Starting Nmap 7.80 ( https://nmap.org ) at 2026-07-27 15:42 IST
Nmap scan report for 192.168.3.36
Host is up (0.0025s latency).
Not shown: 990 filtered ports
PORT     STATE SERVICE
80/tcp   open  http
1801/tcp open  msmq
2103/tcp open  zephyr-clt
2105/tcp open  eklogin
2107/tcp open  msmq-mgmt
3306/tcp open  mysql
6002/tcp open  X11:2
7001/tcp open  afs3-callback
7002/tcp open  afs3-prserver
9000/tcp open  cslistener

Nmap scan report for 192.168.3.146
Host is up (0.00098s latency).
Not shown: 992 closed ports
PORT     STATE SERVICE
135/tcp  open  msrpc
139/tcp  open  netbios-ssn
445/tcp  open  microsoft-ds
5432/tcp open  postgresql
6002/tcp open  X11:2
```

2. Ping Scan

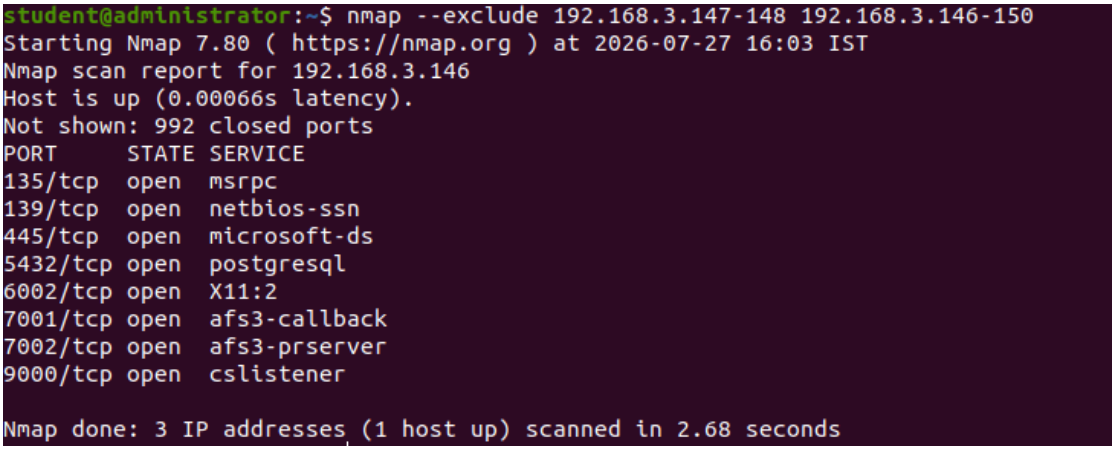
nmap -sP 192.168.3.0/24

```
student@administrator:~$ nmap -sP 192.168.3.0/24
Starting Nmap 7.80 ( https://nmap.org ) at 2026-07-27 15:54 IST
Nmap scan report for 192.168.3.36
Host is up (0.00066s latency).
Nmap scan report for 192.168.3.146
Host is up (0.0020s latency).
Nmap scan report for 192.168.3.153
Host is up (0.00080s latency).
Nmap scan report for 192.168.3.155
Host is up (0.00076s latency).
Nmap scan report for 192.168.3.156
Host is up (0.00073s latency).
Nmap scan report for 192.168.3.159
Host is up (0.00067s latency).
Nmap scan report for 192.168.3.160
Host is up (0.00065s latency).
Nmap scan report for 192.168.3.161
Host is up (0.00050s latency).
Nmap scan report for 192.168.3.164
Host is up (0.00052s latency).
Nmap scan report for 192.168.3.165
Host is up (0.00050s latency).
Nmap scan report for 192.168.3.166
Host is up (0.00052s latency).
Nmap scan report for 192.168.3.168
Host is up (0.00047s latency).
Nmap scan report for 192.168.3.169
Host is up (0.0013s latency).
Nmap scan report for 192.168.3.171
Host is up (0.00042s latency).
Nmap scan report for 192.168.3.172
Host is up (0.0011s latency).
Nmap scan report for 192.168.3.173
Host is up (0.0011s latency).
Nmap scan report for 192.168.3.174
Host is up (0.0011s latency).
Nmap scan report for 192.168.3.177
Host is up (0.0010s latency).
Nmap scan report for 192.168.3.178
Host is up (0.00098s latency).
Nmap scan report for 192.168.3.179
Host is up (0.00029s latency).
Nmap scan report for 192.168.3.180
Host is up (0.00038s latency).
Nmap scan report for 192.168.3.184
Host is up (0.00075s latency).
Nmap scan report for 192.168.3.185
Host is up (0.00073s latency).
Nmap scan report for 192.168.3.186
Host is up (0.00071s latency).
Nmap scan report for 192.168.3.187
```



3. Excluding hosts/networks

nmap --exclude 192.168.3.147-148 192.168.3.146-150



4. To scan open ports

nmap --open mu.ac.in

nmap --open 192.168.3.145

```
student@administrator:~$ nmap --open mu.ac.in
Starting Nmap 7.80 ( https://nmap.org ) at 2026-07-27 16:05 IST
Nmap scan report for mu.ac.in (49.249.201.174)
Host is up (0.00069s latency).
Other addresses for mu.ac.in (not scanned): 64:ff9b::31f9:c9ae
rDNS record for 49.249.201.174: static-174.201.249.49-tataidc.co.in
Not shown: 998 filtered ports
Some closed ports may be reported as filtered due to --defeat-rst-ratelimit
PORT    STATE SERVICE
80/tcp  open  http
443/tcp open  https

Nmap done: 1 IP address (1 host up) scanned in 4.54 seconds
student@administrator:~$ nmap --open 192.168.3.145
Starting Nmap 7.80 ( https://nmap.org ) at 2026-07-27 16:06 IST
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 3.02 seconds
```

5. Show host interfaces and routes

nmap –iflist 192.168.3.145

```
student@administrator:~$ nmap -iflist 192.168.3.145
Starting Nmap 7.80 ( https://nmap.org ) at 2026-07-27 16:07 IST
************************INTERFACES************************
DEV    (SHORT)  IP/MASK                     TYPE     UP MTU   MAC
lo     (lo)     127.0.0.1/8                 loopback up 65536
lo     (lo)     ::1/128                     loopback up 65536
enp2s0 (enp2s0) 192.168.3.194/24            ethernet up 1500  D8:5E:D3:9D:F4:8F
enp2s0 (enp2s0) fe80::86fe:d574:8db3:436b/64 ethernet up 1500  D8:5E:D3:9D:F4:8F

**************************ROUTES**************************
DST/MASK                    DEV    METRIC GATEWAY
192.168.3.0/24              enp2s0 100
169.254.0.0/16              enp2s0 1000
0.0.0.0/0                   enp2s0 100    192.168.3.1
::1/128                     lo     0
fe80::86fe:d574:8db3:436b/128 enp2s0 0
::1/128                     lo     256
fe80::/64                   enp2s0 100
ff00::/8                    enp2s0 256
```

6. Scan specific ports

nmap -p 80 192.168.3.145

nmap -p 80,21 192.168.3.145

```
student@administrator:~$ nmap -p 80 192.168.3.145
Starting Nmap 7.80 ( https://nmap.org ) at 2026-07-27 16:09 IST
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 3.02 seconds
student@administrator:~$ nmap -p 80,21 192.168.3.145
Starting Nmap 7.80 ( https://nmap.org ) at 2026-07-27 16:09 IST
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 3.02 seconds
```

7. IP protocol scan

nmap -sO 192.168.3.145

```
vboxuser@ubuntu:~$ sudo nmap -sO 192.168.3.145
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-07-27 13:51 UTC
Nmap scan report for 192.168.3.145
Host is up (0.00067s latency).
Not shown: 255 open|filtered n/a protocols (no-response)
PROTOCOL STATE SERVICE
6        open  tcp

Nmap done: 1 IP address (1 host up) scanned in 4.82 seconds
```

8. OS Fingerprinting

nmap -O 192.168.3.145

```
vboxuser@ubuntu:~$ sudo nmap -O 192.168.3.145
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-07-27 13:48 UTC
Nmap scan report for 192.168.3.145
Host is up (0.0031s latency).
All 1000 scanned ports on 192.168.3.145 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
Warning: OSScan results may be unreliable because we could not find at least 1 open
and 1 closed port
Aggressive OS guesses: 3Com 4500G switch (92%), H3C Comware 5.20 (92%), Huawei VRP 8
.100 (92%), Microsoft Windows Server 2003 SP1 (92%), Oracle Virtualbox (92%), QEMU u
ser mode network gateway (92%), AXIS 2100 Network Camera (92%), D-Link DP-300U, DP-G
310, or Hamlet HPS01UU print server (92%), HP Tru64 UNIX 5.1A (92%), Sanyo PLC-XU88
digital video projector (92%)
No exact OS matches for host (test conditions non-ideal).

OS detection performed. Please report any incorrect results at https://nmap.org/subm
it/ .
Nmap done: 1 IP address (1 host up) scanned in 6.69 seconds
```

9. Version Detection

nmap -sV 192.168.3.145

```
student@administrator:~$ nmap -sV 192.168.3.145
Starting Nmap 7.80 ( https://nmap.org ) at 2026-07-27 16:12 IST
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 3.24 seconds
```

10. TCP port scan

nmap -sS 192.168.3.145

```
vboxuser@ubuntu:~$ sudo nmap -sS 192.168.3.145
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-07-27 13:52 UTC
Nmap scan report for 192.168.3.145
Host is up (0.0015s latency).
All 1000 scanned ports on 192.168.3.145 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)

Nmap done: 1 IP address (1 host up) scanned in 4.17 seconds
```

11. UDP port scan

nmap -sU 192.168.3.145

```
vboxuser@ubuntu:~$ sudo nmap -sU 192.168.3.145
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-07-27 14:01 UTC
Nmap scan report for 192.168.3.145
Host is up (0.00037s latency).
All 1000 scanned ports on 192.168.3.145 are in ignored states.
Not shown: 1000 open|filtered udp ports (no-response)

Nmap done: 1 IP address (1 host up) scanned in 1600.00 seconds
```

12. Fast Scan

nmap -f 192.168.3.145

```
vboxuser@ubuntu:~$ sudo nmap -f 192.168.3.145
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-07-27 13:55 UTC
Nmap scan report for 192.168.3.145
Host is up (0.0018s latency).
All 1000 scanned ports on 192.168.3.145 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)

Nmap done: 1 IP address (1 host up) scanned in 4.33 seconds
```

13. Find out if a host/network is protected by a firewall

nmap -sA mu.ac.in

```
vboxuser@ubuntu:~$ sudo nmap -sA mu.ac.in
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-07-27 13:56 UTC
Nmap scan report for mu.ac.in (49.249.201.174)
Host is up (0.00044s latency).
rDNS record for 49.249.201.174: static-174.201.249.49-tataidc.co.in
All 1000 scanned ports on mu.ac.in (49.249.201.174) are in ignored states.
Not shown: 1000 unfiltered tcp ports (reset)

Nmap done: 1 IP address (1 host up) scanned in 0.44 seconds
```

14. **Scan a host when protected by the firewall**

nmap -PN mu.ac.in

```
administrator@administrator:~$ nmap -PN mu.ac.in
Starting Nmap 7.80 ( https://nmap.org ) at 2026-07-27 16:42 IST
Nmap scan report for mu.ac.in (49.249.201.174)
Host is up (0.00057s latency).
rDNS record for 49.249.201.174: static-174.201.249.49-tataidc.co.in
Not shown: 998 filtered ports
PORT    STATE SERVICE
80/tcp  open  http
443/tcp open  https

Nmap done: 1 IP address (1 host up) scanned in 4.83 seconds
```

15. **The fastest way to scan all your devices/computers for open ports ever**

nmap -T5 mu.ac.in

```
administrator@administrator:~$ nmap -T5 mu.ac.in
Starting Nmap 7.80 ( https://nmap.org ) at 2026-07-27 16:39 IST
Nmap scan report for mu.ac.in (49.249.201.174)
Host is up (0.0010s latency).
rDNS record for 49.249.201.174: static-174.201.249.49-tataidc.co.in
Not shown: 998 filtered ports
PORT    STATE SERVICE
80/tcp  open  http
443/tcp open  https

Nmap done: 1 IP address (1 host up) scanned in 3.04 seconds
```

Nmap done: 1 IP address (1 host up) scanned in 12.14 seconds

16. **To run an ARP ping scan,** type the following command into the command line:

nmap -sP 192.100.1.1/24

```
vboxuser@ubuntu:~$ sudo nmap -sP 10.0.2.15/24
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-07-27 14:00 UTC
Nmap scan report for _gateway (10.0.2.2)
Host is up (0.00046s latency).
MAC Address: 52:55:0A:00:02:02 (Unknown)
Nmap scan report for 10.0.2.3
Host is up (0.00039s latency).
MAC Address: 52:55:0A:00:02:03 (Unknown)
Nmap scan report for ubuntu (10.0.2.15)
Host is up.
Nmap done: 256 IP addresses (3 hosts up) scanned in 2.06 seconds
```

This will return a list of hosts that responded to your ping requests along with a total number of IP addresses at the end.

Conclusion: Thus we have successfully studied various options of nmap